Youtube Video Preformance Predictor

In [2]:
```
# import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

In [3]:
```
# prepare for the data
data = {
    'ctr': [3.2, 5.8, 2.1, 7.4, 4.5, 6.2, 1.8, 5.1, 3.9, 8.5,
            4.2, 2.8, 6.8, 3.5, 7.1, 2.4, 5.5, 4.8, 6.5, 3.1,
            7.8, 2.6, 5.3, 4.1, 6.9, 3.7, 8.1, 2.2, 5.9, 4.6],

    'total_views': [12000, 28000, 8500, 42000, 19000, 33000, 7000, 24000, 16000, 51000,
                    18000, 11000, 38000, 14500, 44000, 9500, 26000, 21000, 35000, 13000,
                    47000, 10000, 25000, 17500, 40000, 15000, 49000, 8000, 31000, 20000]
}
```

In [5]:
```
df = pd.DataFrame(data)
df.head()
```

Out[5]:

	ctr	total_views
0	3.2	12000
1	5.8	28000
2	2.1	8500
3	7.4	42000
4	4.5	19000

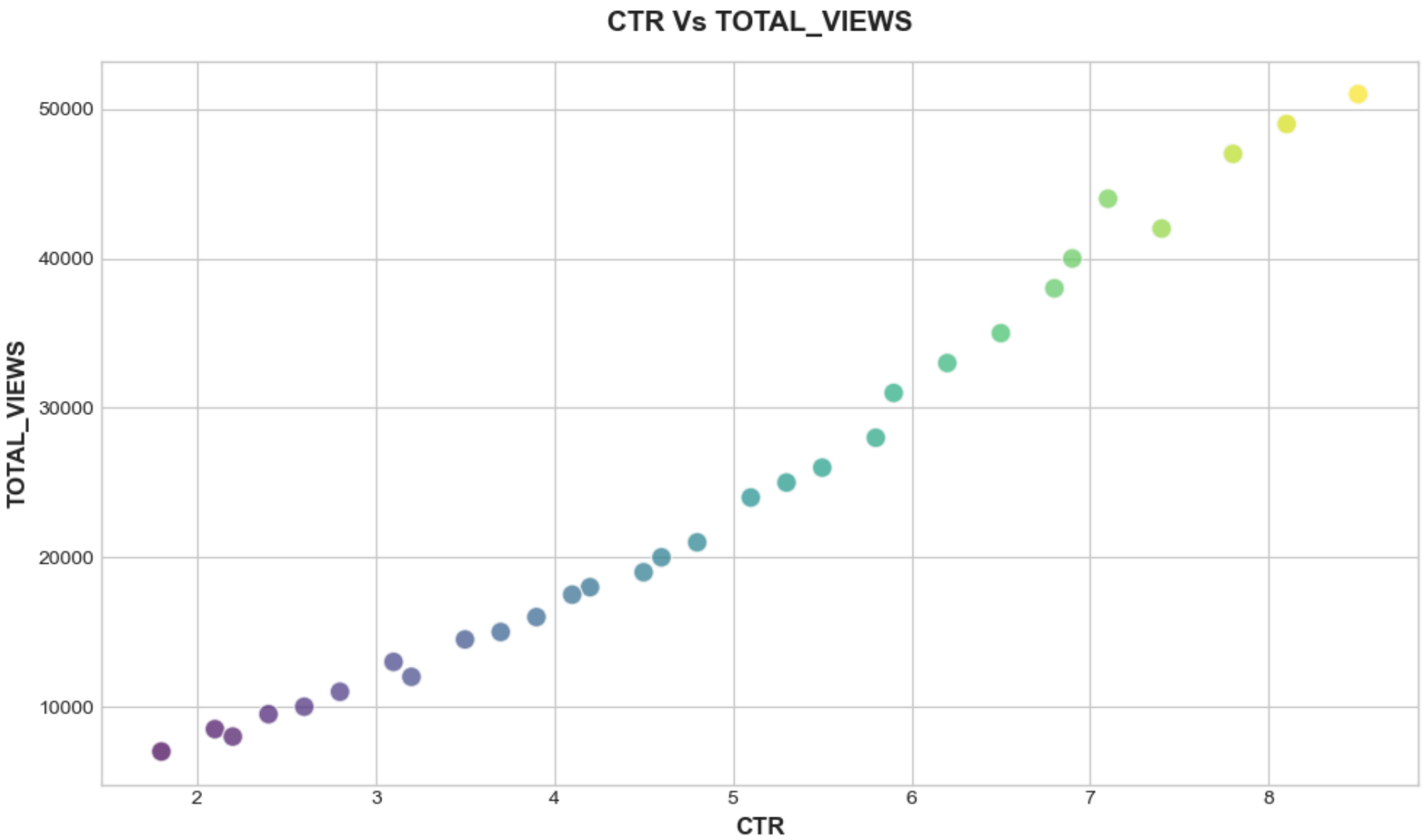
In [23]:
```
df.describe()
```

Out[23]:

	ctr	total_views
count	30.000000	30.000000
mean	4.880000	24433.333333
std	1.943406	13537.134962
min	1.800000	7000.000000
25%	3.275000	13375.000000
50%	4.700000	20500.000000
75%	6.425000	34500.000000
max	8.500000	51000.000000

In [59]:

```
# Visualize the given data
plt.style.use('seaborn-v0_8-whitegrid')
plt.figure(figsize=(10,6))
plt.scatter(df['ctr'],df['total_views'],c=df['ctr'], cmap='viridis', s=100, alpha=0.7, edgecolors='white')
plt.xlabel('CTR',fontsize=12, fontweight='bold')
plt.ylabel('TOTAL_VIEWS',fontsize=12, fontweight='bold')
plt.title("CTR Vs TOTAL_VIEWS",fontsize=14, fontweight='bold', pad = 15)
plt.tight_layout()
plt.show()
```



In [38]:
```
# prepare data for training
X = df[['ctr']] # sklearn expects 2-D array
y = df['total_views']
```

In [39]:
```
# splitting into training and testing data
random_state = 42
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size= 0.2)
```

In [40]:
```
# create the model and train it
model = LinearRegression()
model.fit(X,y)
```

Out[40]:
```
▾ LinearRegression
▶ Parameters
```

In [57]:
```
print("\nModel Parameters:")
print(f"Slope (Coefficient): {model.coef_[0]:.2f}")
print(f"Intercept: {model.intercept_:.2f}")
```

```
Model Parameters:
Slope (Coefficient): 6879.61
Intercept: -9139.17
```

In [42]:
```
y_test
```

Out[42]:
```
0     12000
14    44000
5     33000
4     19000
26    49000
8     16000
Name: total_views, dtype: int64
```

In [45]:
```
# Make predictions on test data
y_pred = model.predict(X_test)
y_pred
```

Out[45]:
```
array([12875.58736883, 39706.06907214, 33514.4194483 , 21819.08126993,
       46585.6797653 , 17691.31485404])
```

In [47]:
```
results = pd.DataFrame({'Actual': y_test.values, 'Predicted': y_pred.round()})
```

```
results
```

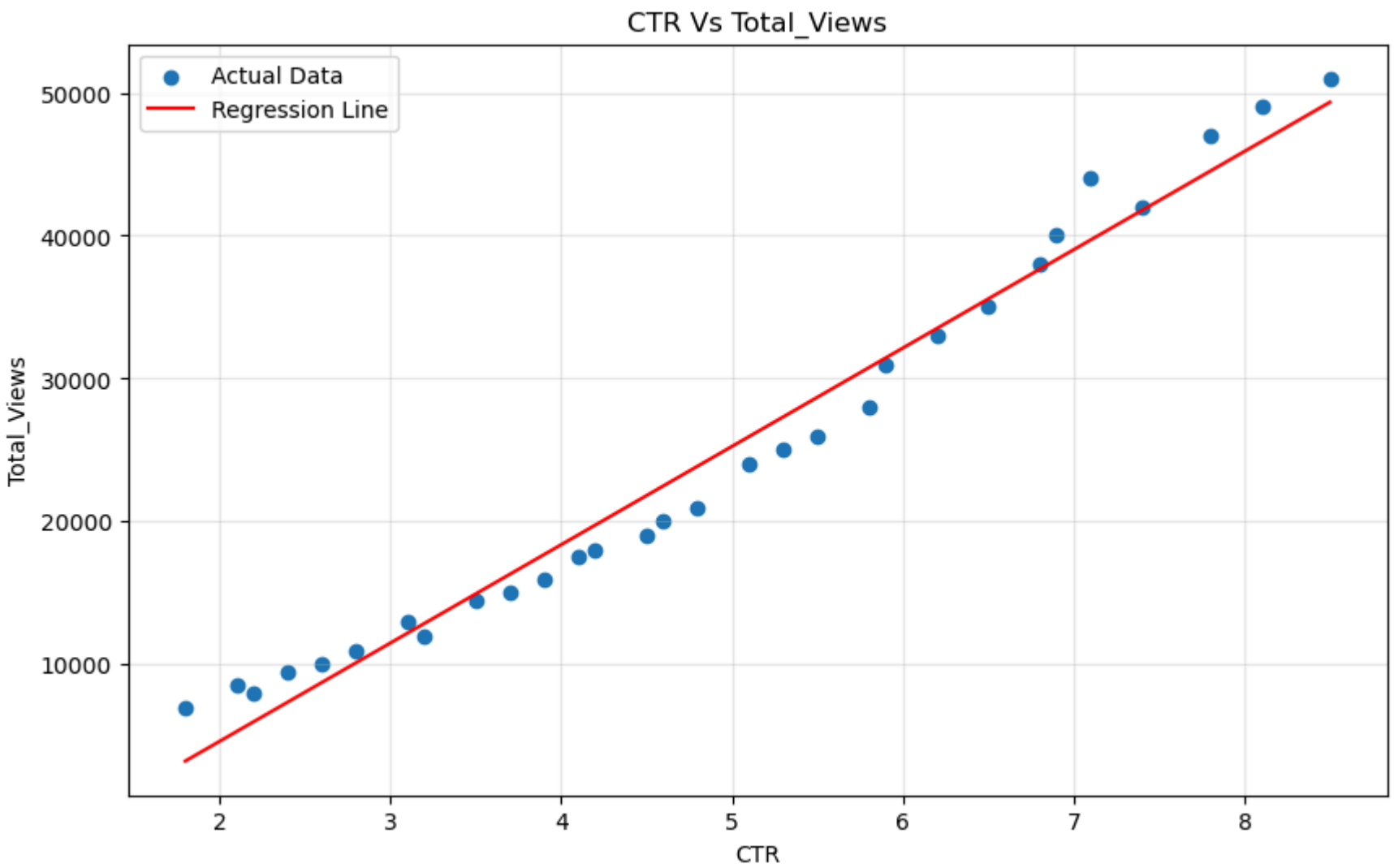
Out[47]:

	Actual	Predicted
0	12000	12876.0
1	44000	39706.0
2	33000	33514.0
3	19000	21819.0
4	49000	46586.0
5	16000	17691.0

In [50]:

```
# Visualize the Regression Line
plt.figure(figsize = (10,6))
plt.scatter(df['ctr'],df['total_views'], label= 'Actual Data')
X_line = np.linspace(df['ctr'].min(), df['ctr'].max(),100).reshape(-1,1)
'''we use linspace because the data what we had was not continuous which doesnot give the best regression line. so we need to
min and max value form the given data and make more data which is 100 values so that we can expect the best regression line an
to 2-D for the prediction of those values '''
plt.plot(X_line, model.predict(X_line), color = 'red', label= 'Regression Line')
plt.xlabel('CTR')
plt.ylabel('Total_Views')
plt.title('CTR Vs Total_Views')
plt.grid(True, alpha = 0.3)
plt.legend()
plt.show()
```

```
C:\Users\G.Guna\desktop\ML\heart_disease_project\env\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does no
t have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```



In [53]:
```
new_ctr = 8
new_pred_views = model.predict([[new_ctr]])
new_pred_views
print(f"\nArjun's Question: If I get {new_ctr}% CTR, how many views can i expect ?")
print(f"Predicted Total Views: {new_pred_views[0]:,.0f}")
```

```
Arjun's Question: If I get 8% CTR, how many views can i expect ?
Predicted Total Views: 45,898
```

```
C:\Users\G.Guna\desktop\ML\heart_disease_project\env\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does no
t have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

In [ ]: